Elapsed time is the reward model to beat: failure judges and monitors on a low-cost arm

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Abstract

Watching a robot run only up to half the length of the task's longest run, elapsed time already separates failed from successful runs with AUROC 0.851 within policy (CI 0.829–0.873) on 2,047 public SO-101 rollouts from 7 policies: failed runs are slow, not timed out (only 8.7 % reach the task's longest length). The dataset's stopping rule decides what a reward model can show: where runs end at a time limit, elapsed time is the reward model to beat; where an operator stops them, length is blind and judges work. Over the whole run, length gives 0.938 pooled and 0.942 within policy, above two zero-shot reward models proposed as failure detectors: Robometer-4B (0.818 pooled) and TOPReward with an 8B video-language model (0.661 within policy). On 88 of our own rollouts, stopped by an operator, length carries nothing (0.52) and TOPReward reaches 0.82 on 69 of them. There, a deterministic top-camera check at rest makes no false positive, a robot-trained 27B judge catches 8 of 8 failures at 2 of 42 false alarms, and its base model does almost as well given the same input; every mid-episode monitor we tested misses the bar. Report a length baseline, within policy.

1 Introduction

A reward model that scores robot rollouts is also a failure detector, and papers evaluate it as one. None of the reward-model papers in our library reports how far plain episode length gets on the same data. We measure it on a public SO-101 benchmark and on our own rollouts, with the analyses frozen before the numbers, and set every learned judge and monitor we ran against it.

2 Data and labels

ArmnetBench v0.1 (SO-101, 8 tasks, 7 policies): 2,047 policy rollouts with a success or failure class (515 successes, 1,532 failures); teleoperated episodes excluded. Our rollouts: 88 labelled bowls-task episodes on one SO-101 and a Jetson Orin NX, and the 50-episode subset (42 successes, 8 failures) every monitor gate used. Labels are the operator's, not blind to policy.

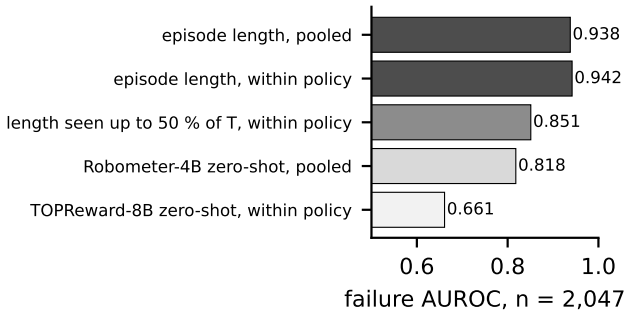


Figure 1: Failure AUROC on ArmnetBench: episode length vs two zero-shot reward models (Robometer-4B, TOPReward-8B).

3 The length baseline

Pooled over policies, length gives 0.938; Robometer-4B, run zero-shot as its authors ship it, 0.818 (CI 0.790–0.843); TOPReward (Qwen3-VL-8B, no training, the paper's success score) 0.653 pooled and 0.661 within policy (CI 0.631–0.692), below length by -0.31 to -0.25 (paired 95 % CI; protocol frozen before any score). Within each policy, length ranges 0.91–0.96, pair-weighted 0.942 (CI 0.927–0.956): it is not the policy identity (Fig. 1).

4 Is it the timeout?

Only 8.7 % of failures run to the task's longest episode. Observing length only up to half of that, when 88 % of failures but 19 % of successes are still running, gives 0.851 within policy (CI 0.829–0.873; Fig. 2; TOPReward on the same cut clips: 0.667): failed rollouts are slow, not timed out. Both pre-registered readings hold: length is a within-policy null, and it is more than the timeout. On our own operator-stopped rollouts the same baseline is at chance (0.52), Robometer reaches 0.577, and TOPReward 0.82 on 69 of them (CI 0.70–0.93; policy identity alone 0.87): the baseline depends on how episodes end, which is why it must be reported per dataset.

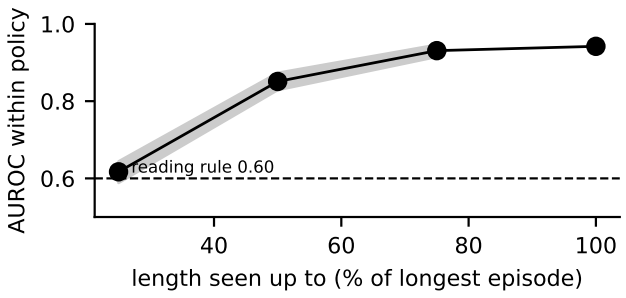


Figure 2: Length observed up to a fraction of the task's longest episode, AUROC within policy (band: 95 % CI).

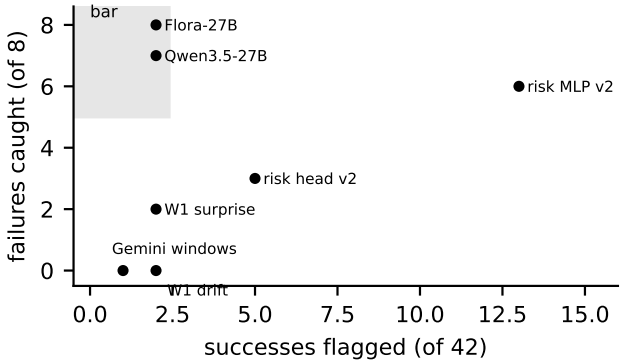


Figure 3: Monitors on 50 own rollouts: failures caught (of 8) vs successes flagged (of 42); shaded: the bar.

5 Judges and monitors on our rollouts

At rest. A deterministic top-camera check (colour masks and geometry, no learning) is right on 94.3 % of judged episodes with 0 false positives (Wilson 0–13 two-view clip and NVIDIA's prompt, Flora-Rewarder-27B flags 8 / 8 failures at 2 / 42 successes (AUROC 1.00; Wilson on the false alarms 1–16 at 2 / 42 on the same input; a 9B model (0.75) and Gemini 3.8 Flash (0.96, one false alarm over the bar) do not pass. The input format, not the robot fine-tune, decided the result.

Mid-episode. A Gemini sliding-window judge catches 0 / 8; the surprise of a separate world model 2 / 8 (AUROC 0.86, Wilson 7–59 the world model's latent reached 0.93 in a first version that read elapsed time; the preregistered second version, without time or length, catches 3 / 8 at 5 / 42 (AUROC 0.79 vs 0.71 with shuffled actions; Wilson 14–69 ($\geq$ 5 / 8 at $\leq$ 1 / 20), and the first version's result is withdrawn (Fig. 3).

6 Limitations

Two task families (ArmnetBench's eight tasks, our bowls task); our labels are the operator's and not blind; the risk head is trained and tested on the same 50 episodes (5-fold by episode: episodes do not leak, tasks and policies do); Robometer under the within-policy and truncation analyses is pending its per-episode scores; TOPReward's prefix count, frames per query and Yes/No wording are not given in its paper and are ours; the timeout is approximated by each task's longest episode.

References

Robometer 2603.02115 · SARM 2509.25358 · TOPReward 2602.19313 · VLLR 2604.00055 · SARM2 2606.10305 · STDR 2606.31377 · No Free Checker 2609.09250 · FARM 2609.11445 · ReAP 2610.16309 · SAFE 2506.09937 · FAIL-Detect 2503.08558 · VLA-Corrector 2609.06508 · Spotter 2609.36808 · REMORA 2610.39125 · ArmnetBench 2607.24481 · SO-101 benchmark 2606.08881.